Augmenting Microloan Risk Modeling through Data Synthesis and Transfer Learning

A Capstone Defense

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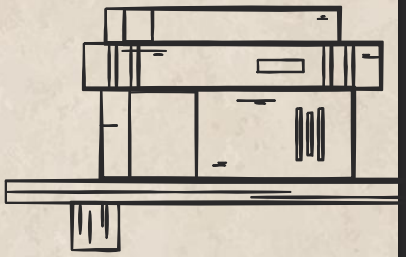


01

Introduction

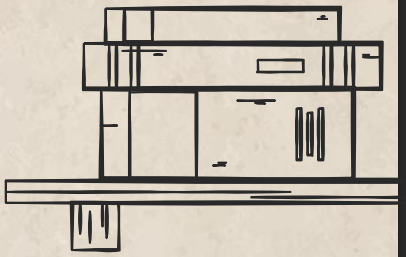
Introduction

In emerging economies, financial exclusion driven by information asymmetry prevents millions from accessing formal credit.

Deep Learning and ANNs have transformed financial decision-making, yet most models rely on data from developed nations and often fail in developing regions due to informal employment, economic gaps, and sparse credit histories.

This research proposes a deep learning model using Transfer Learning that pretrains an ANN on a US bank loan dataset and transfers learned patterns to Kiva microloan data, combining financial and socioeconomic factors into a unified risk scoring framework.

Explainable AI techniques such as SHAP and LIME are applied to ensure transparency and fairness.



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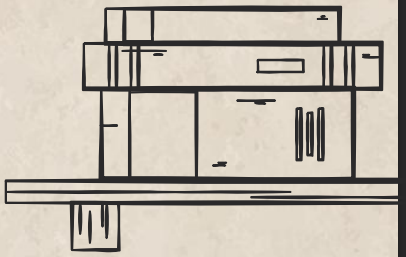
Literature Review

Literature Review

Reference	Objective	Methodology	Outcome
Alagic et al. [1]	Enhance credit risk analysis with mental health data integration.	Ensemble classifier (RF, XGBoost, ADABoost, LightGBM) on German and Australian datasets.	Improved accuracy of 85% with interpretability. Training complexity noted limitations.
Condori-Alejo et al. [2]	Rural micro-credit assessment in a Peruvian microfinance institution.	LightGBM with data asymmetry correction, income and repayment history assessed.	96.2% accuracy in rural microfinance credit risk assessment.
Nallakaruppan et al. [3]	Credit risk assessment, financial decision support XAI.	Logistic Regression integrated with Random Forest and explainable AI. Feature-level explanations applied.	93% predictive accuracy, enhanced transparency through XAI feature explanations.
Mamun et al. [4]	Predict bank loan eligibility using ML models	Comparison-based study on Kaggle dataset(10,128 instances); Logistic Regression applied.	Logistic Regression achieved 92% accuracy, data imbalance and dynamism noted.

The current literature prioritizes predictive accuracy over contextual explainability



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Problem Statements

Problem Statements

Data Sparsity

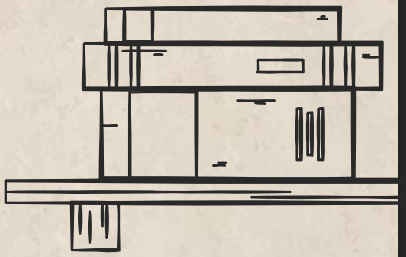
Models trained primarily on developed-nation data cannot comprehend incomplete paperwork and financial behaviours.

Algorithmic Rejection

Creditworthy borrowers are systematically denied simply because they do not fit formal Western credit profiles.

The Black Box Trust Deficit

Opaque Machine Learning decisions generate mistrust among both financial institutions and local borrowers.



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Motivation

Motivation

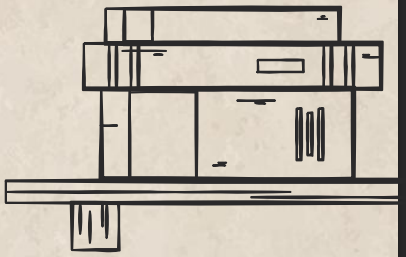
Bank loans are vital for economic growth, but approval rates differ significantly between developed nations (40%–60% for personal loans, 50%–70% for SME loans) and developing nations,

20% - 40%
Personal Loan Approvals

30% - 50%
SME Loan Approvals

Although microloans have the greatest potential for community economic development, they lack the sophisticated analytical frameworks required for approval.

This research seeks to address these differences by developing inclusive credit assessment methods that consider informal income sources and non-traditional credit histories, promoting unbiased loan access for small businesses and rural communities.



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Objectives

Objectives

1 Model Stability

Build Artificial Neural Networks capable of high-precision prediction on multi-source datasets.

2 Contextual Intelligence

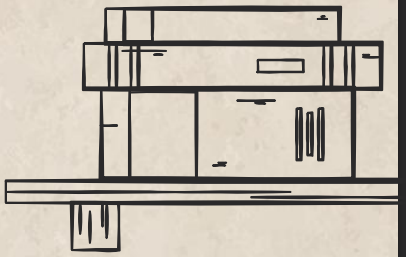
Synthesize Kiva microloan data with regional Multi-Dimensional Poverty Index (MPI) and macroeconomic profiles.

3 Cross-Regional Transfer

Shift predictive intelligence from data-rich US banks to data-poor microfinance ecosystems.

4 Algorithmic Transparency

Integrate Explainable AI (LIME & SHAP) to guarantee fairness, accountability, and intelligibility.



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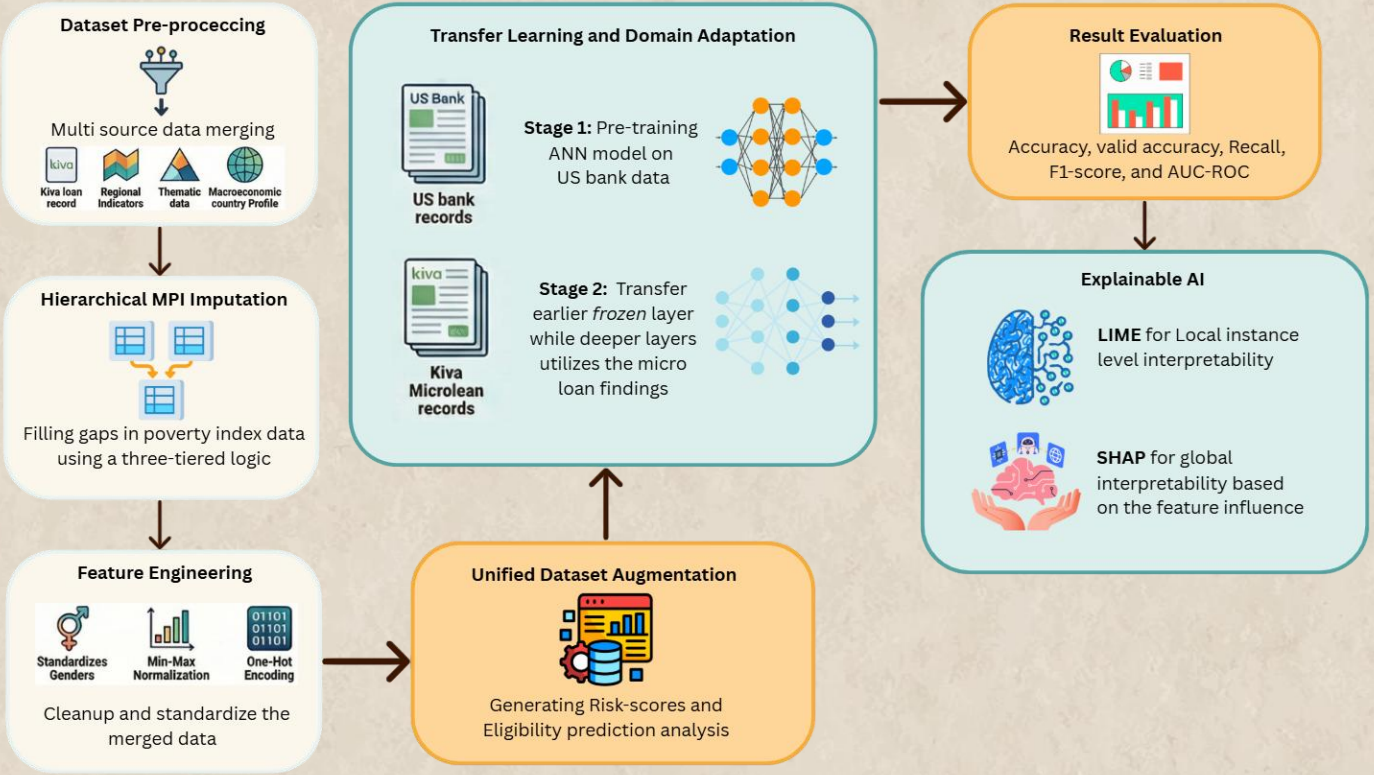
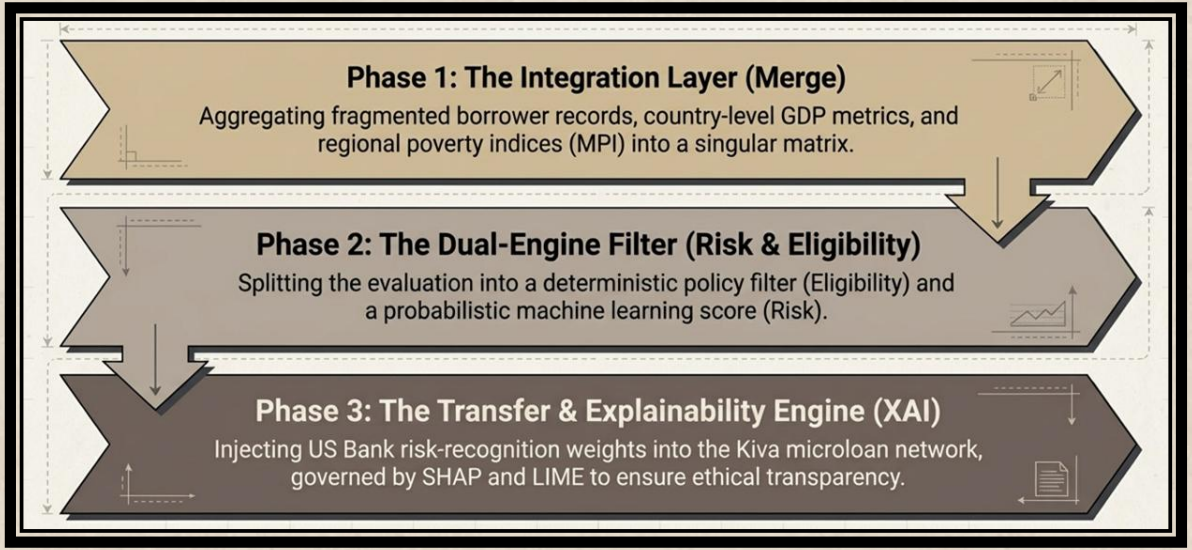
Methodology

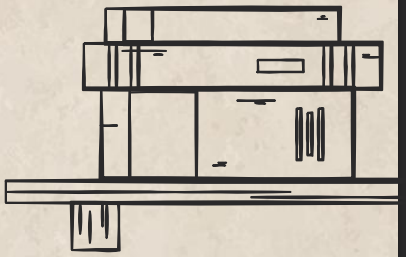
Proposed Model



Model Architecture

The proposed model uses a three-step architecture, which moves the data fusion step to a unified evaluation step, and then to an Explainable Transfer Learning engine.





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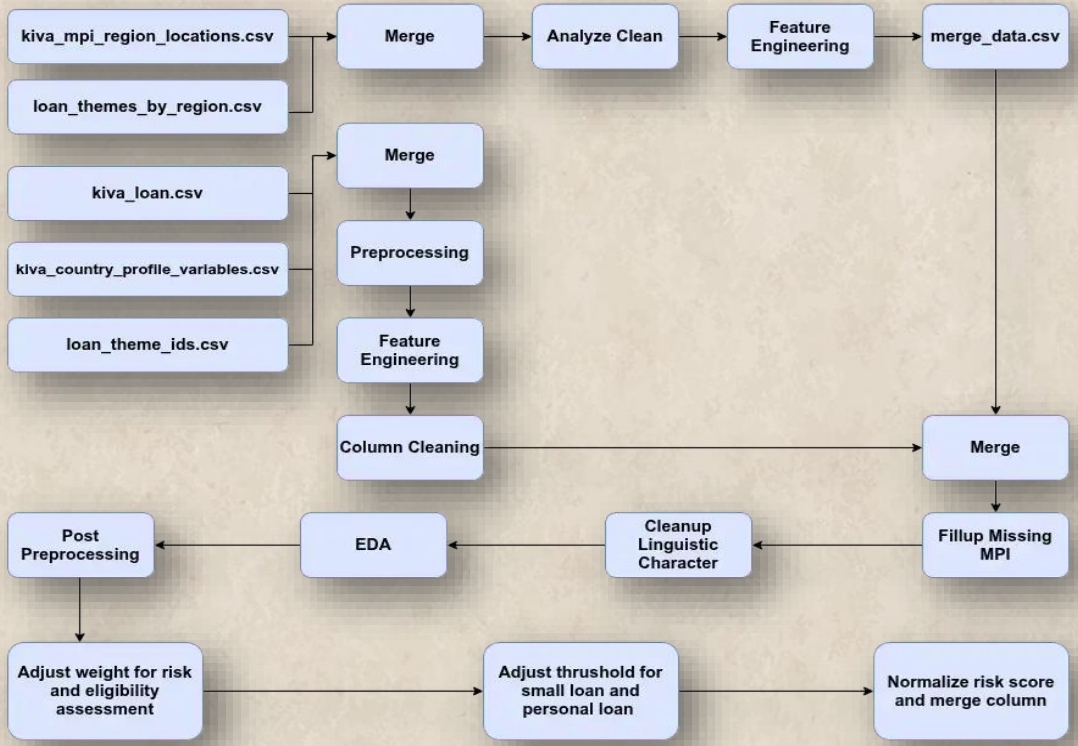
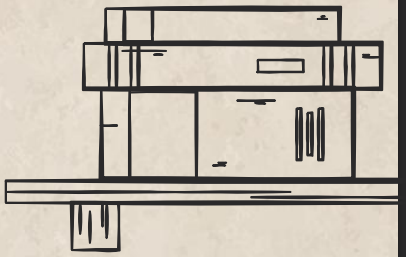
Dataset Augmentation

Datasets

Datasets	Records	**Columns**	**Role**
1. US Bank Loan Dataset	20,000	37	Source Domain
2. Kiva Country Profile	86	38	Macro-economic Context
3. Kiva Loan Theme by Region	15,763	16	Regional Integration
4. Kiva MPI Region Locations	1009	9	Poverty Index Mapping
5. Kiva Loans (Core)	6,71,205	19	Target Domain

Data Augmentation for a unified Kiva dataset





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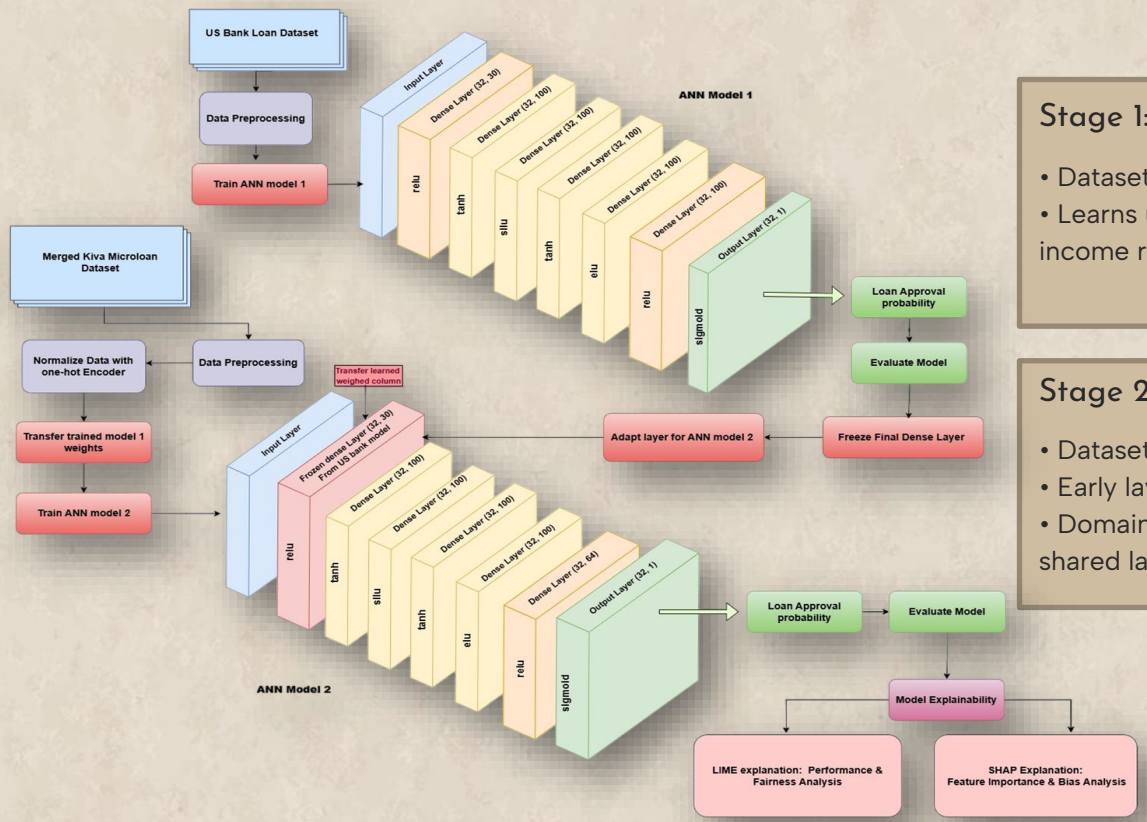
Transfer Learning

Transfer Learning and Domain Adaptation

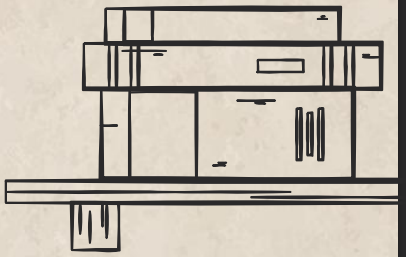


Stage 1: Pre-training Source Domain

- Dataset: US Bank Loan (20,000 records)
- Learns universal credit risk patterns (debt-to-income ratios, payment behaviour)

Stage 2: Transferring to Target Domain

- Dataset: Merged Kiva Microloan dataset
- Early layers FROZEN to retain banking knowledge
- Domain Adaptation: feature-based mapping to shared latent space



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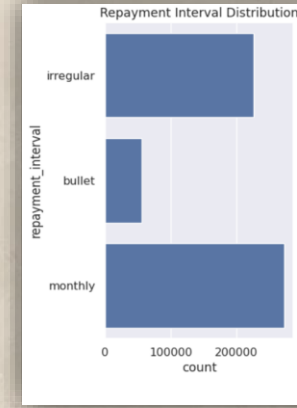
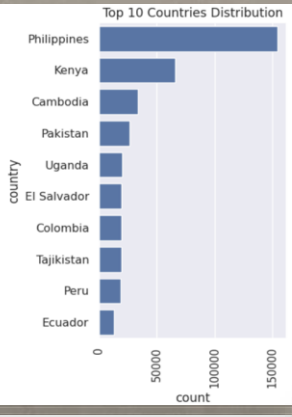
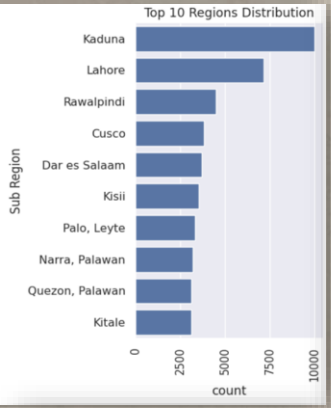
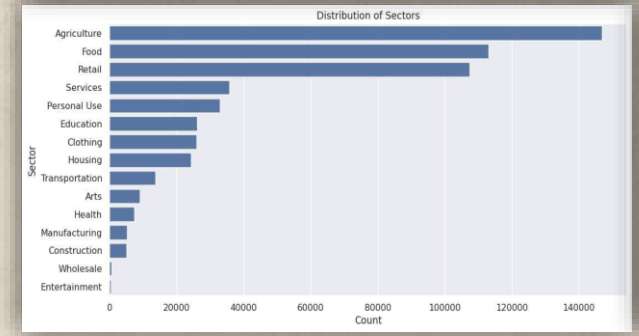
Results & Evaluation

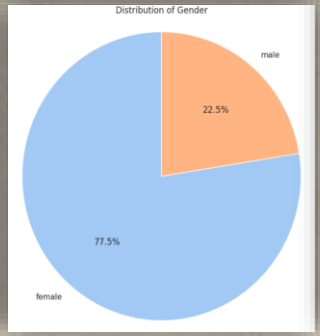
Data Visualization & Correlation

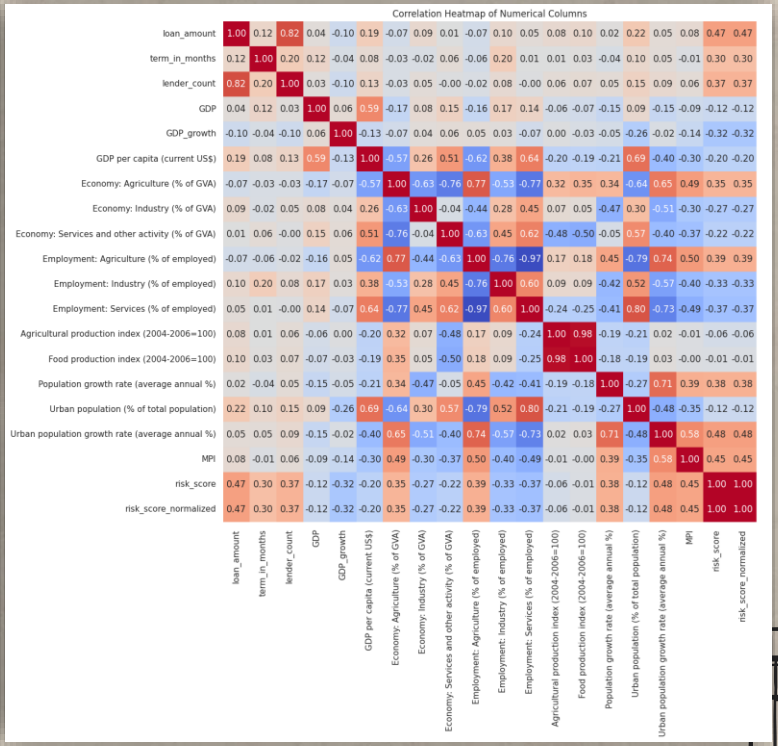




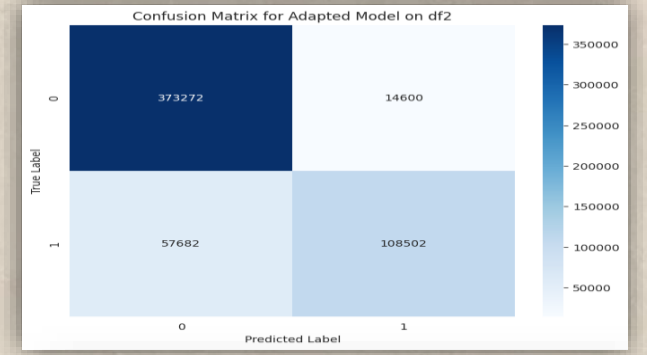


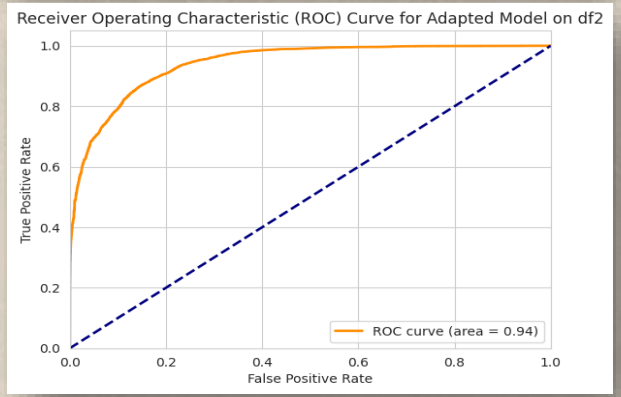






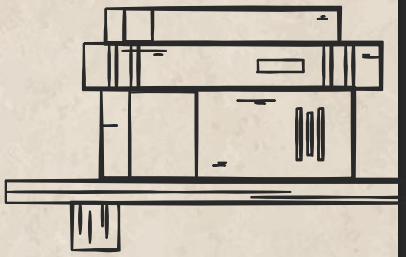
Results – Model Performance





Metric	Baseline (Scratch)	Transfer Learning (Proposed)	Improvement (Δ)
Validation Accuracy	85.24%	89.52%	+4.28%
Overall Accuracy	82.32%	86.95%	+4.63%
AUC	0.84	0.90	+0.06%
F1-Score	0.81	0.86	+0.05%

Transfer learning successfully bridges the data gap, yielding a +4.28% predictive advantage.



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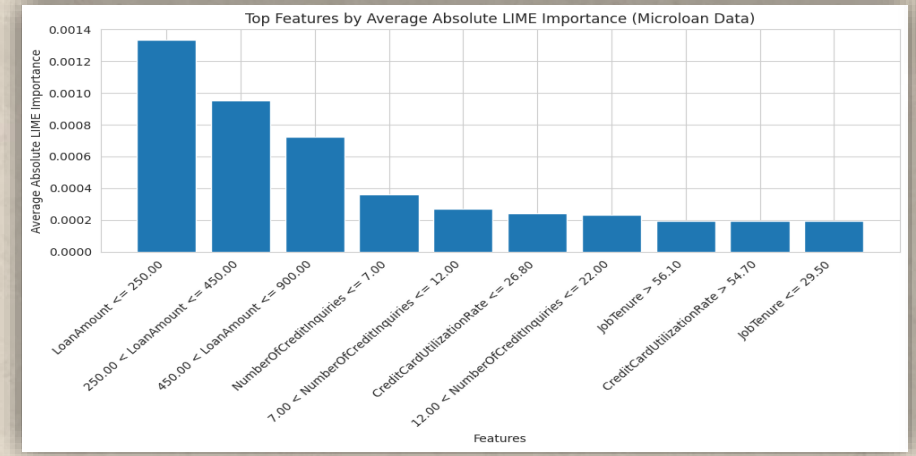
Explainable AI

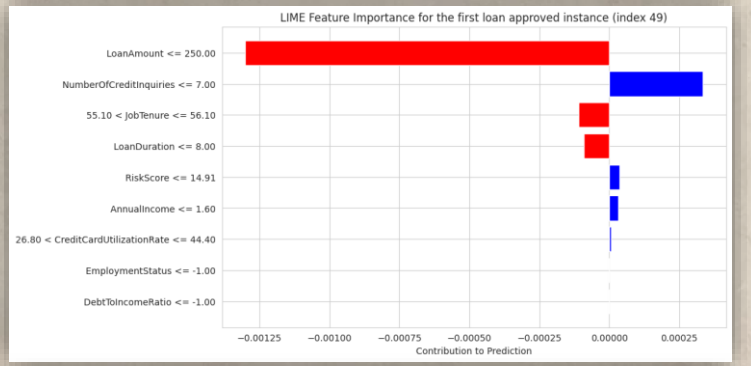
LIME – Local Interpretability

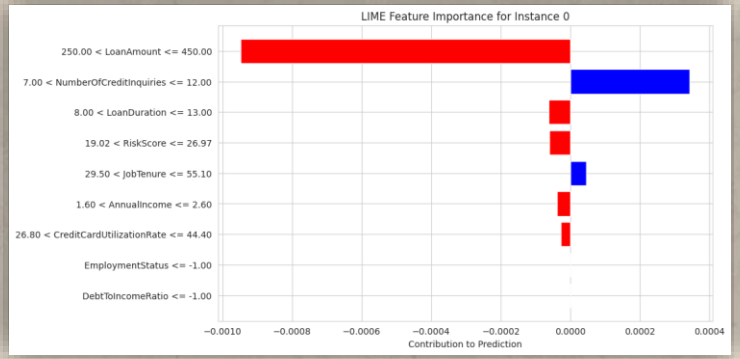
LIME: Provides granular and instance-level insight

Non-Approved: Annual Income and Credit Score had strong negative impacts due to lower values, which pushed the model toward rejection.

Approved: The same features contributed positively when the values were strong.

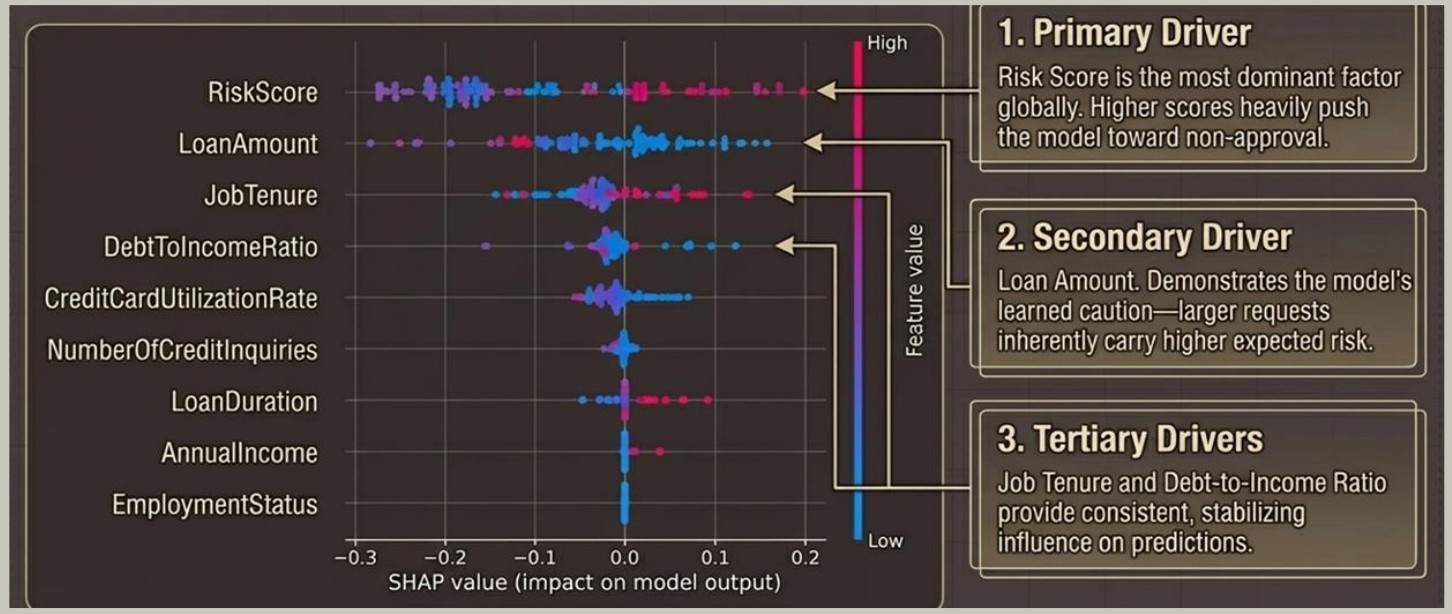






SHAP – Global Interpretability

SHAP measures each feature's contribution to predictions.



1. Primary Driver

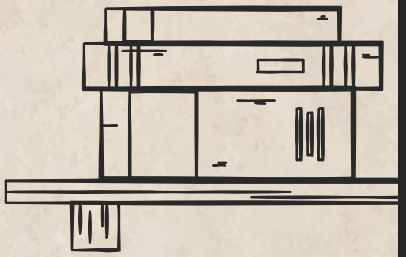
Risk Score is the most dominant factor globally. Higher scores heavily push the model toward non-approval.

2. Secondary Driver

Loan Amount. Demonstrates the model's learned caution—larger requests inherently carry higher expected risk.

3. Tertiary Drivers

Job Tenure and Debt-to-Income Ratio provide consistent, stabilizing influence on predictions.



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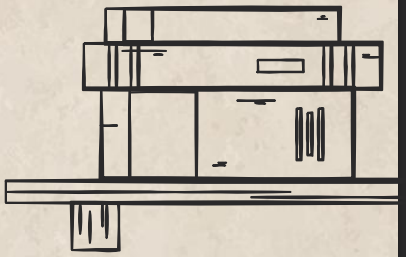
Conclusion

Conclusion

This research successfully developed a three-step computational framework that bridges the gap between data-abundant banking systems and low-resource microfinance settings. By integrating US Bank and Kiva datasets through a rigorous merge pipeline and knowledge transfer, we overcame the data sparsity that commonly restrains credit modelling in developing regions.

The Transfer Learning model achieved 89.52% validation accuracy and 86.95% overall accuracy on the Kiva microloan dataset, outperforming the baseline scratch model by 4.28% and yielding an AUC of 0.90. SHAP analysis confirmed that predictions are driven by economically sound variables, especially Risk Score and Loan Amount.

LIME and SHAP provided instance-level explanations that make individual loan decisions comprehensible. Together, these findings demonstrate that transferring financial intelligence is not only feasible but necessary to make the credit scoring system more reliable, adaptable, fair, and inclusive.



12

Future Scope

Future Scope

Broader Datasets

Incorporate more geographically and temporally heterogeneous datasets to improve the model's ability to generalize across different financial systems.

Deployment Platform

Build a user-friendly application platform capable of delivering transparent and up-to-date loan credit assessment to financial institutions in real time.

Automated Risk Weights

Use sophisticated optimization or attention-based methods to automatically discover risk-score weightings, minimizing human intervention.

Advanced Architectures

Explore ensemble models, hybrid deep learning, or sequential borrower histories to uncover finer financial patterns.

Thank you!

Please feel free to ask any questions.